

Introduction to Robotics

Universidade Federal de Pernambuco

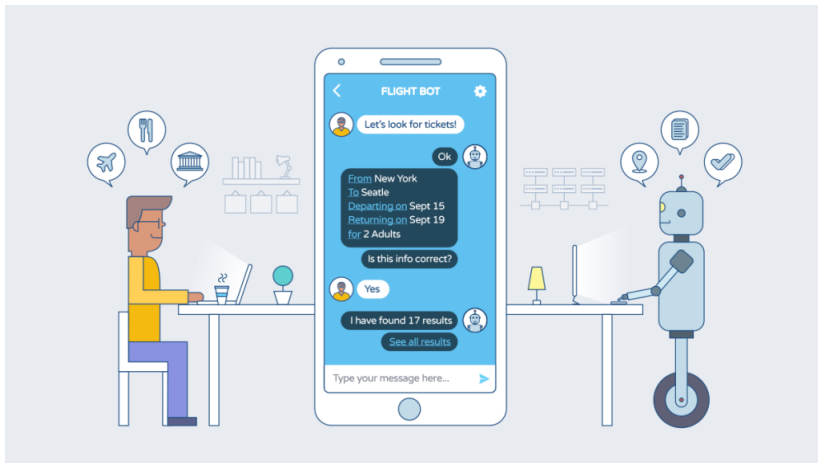
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What is a robot?

- The robot is seen as a machine that, independently of its exterior, is able to modify the environment in which it operates
- This is accomplished by carrying out actions that are conditioned by certain rules of behaviour intrinsic in the machine as well as by some data the robot acquires on its status and on the environment
- In fact, robotics is commonly defined as the science studying the intelligent connection between perception and action

Robot?



Robot?



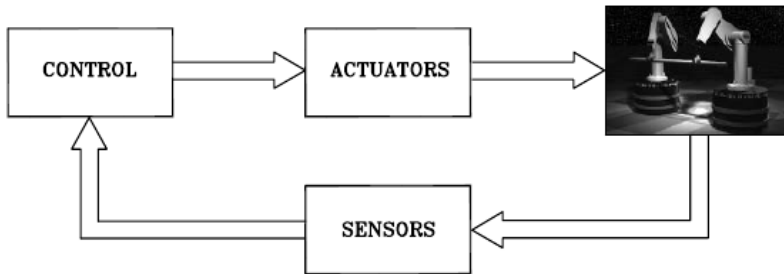
Robot?

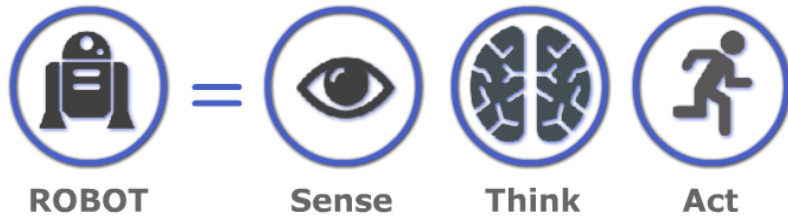


- software (“bots”, AI, Robotic Process Automation -RPA)
- remote-controlled drones, UAV, UGV, UUV
- voice assistants
- autonomous cars
- ATMs, smart washing machines, etc.

A robotic system

- A robotic system is in reality a complex system, functionally represented by multiple subsystems





Mechanical system

- The essential component of a robot is the **mechanical system** endowed, in general, with
 - a locomotion apparatus (wheels, crawlers, mechanical legs)
 - a manipulation apparatus (mechanical arms, end-effectors, artificial hands)
- The realization of such a system refers to the context of design of articulated mechanical systems and choice of materials



Actuation system

- The capability to exert an action, both locomotion and manipulation, is provided by an **actuation system** which animates the mechanical components of the robot
- The concept of such a system refers to the context of **motion control**, dealing with servomotors, drives and transmissions



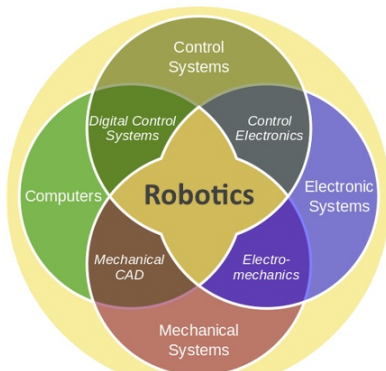
Sensory system

- The capability for perception is entrusted to a **sensory system** which can acquire data on
 - the internal status of the mechanical system (proprioceptive sensors, such as position transducers)
 - the external status of the environment (exteroceptive sensors, such as force sensors and cameras)
- The realization of such a system refers to the context of materials properties, signal conditioning, data processing, and information retrieval



- The capability for connecting action to perception in an intelligent fashion is provided by a **control system** which can command the execution of the action in respect to the goals set by a task planning technique, as well as of the constraints imposed by the robot and the environment
- The realization of such a system follows the same feedback principle devoted to control of human body functions, possibly exploiting the description of the robotic system's components (modelling)
- The context is that of cybernetics, dealing with control and supervision of robot motions, artificial intelligence and expert systems, the computational architecture and programming environment

- Therefore, it can be recognized that robotics is an interdisciplinary subject concerning the cultural areas of
 - mechanics
 - control
 - computers
 - electronics



Applications

Industrial Robotics

- Industrial robotics is the discipline concerning robot design, control and applications in industry
- Its products have by now reached the level of a mature technology.
- The connotation of a robot for industrial applications is that of operating in a structured environment whose geometrical or physical characteristics are mostly known a priori.
- Limited autonomy is required.

- The industrial robot is a machine with significant characteristics of versatility and flexibility.
- According to the widely accepted definition of the Robot Institute of America, a robot is:
A reprogrammable multifunctional manipulator designed to move materials, parts, tools or specialized devices through variable programmed motions for the performance of a variety of tasks
Such a definition, dating back to 1980, reflects the current status of robotics technology.
- Industrial robots present three fundamental capacities that make them useful for a manufacturing process:
 - material handling
 - manipulation
 - measurement

- In a manufacturing process, each object has to be transferred from one location in the factory to another in order to be stored, manufactured, assembled, and packed.
- During transfer, the physical characteristics of the object do not undergo any alteration.
- The robot's capability to pick up an object, move it in space on predefined paths and release it makes the robot itself an ideal candidate for material handling operations.
- Typical applications include:
 - palletizing (placing objects on a pallet in an ordered way),
 - warehouse loading and unloading,
 - mill and machine tool tending,
 - part sorting,
 - packaging.

Video

- In these applications, besides robots, Automated Guided Vehicles (AGV) are utilized which ensure handling of parts and tools around the shop floor from one manufacturing cell to the next.
- As compared to the traditional fixed guide paths for vehicles (inductive guide wire, magnetic tape, or optical visible line), modern AGVs utilize high-tech systems with onboard microprocessors and sensors (laser, odometry, GPS) which allow their localization within the plant layout.

Manufacturing

- Manufacturing consists of transforming objects from raw material into finished products.
- During this process, the part either changes its own physical characteristics as a result of machining, or loses its identity as a result of an assembly of more parts.
- The robot's capability to manipulate both objects and tools make it suitable to be employed in manufacturing.
- Typical applications include:
 - arc and spot welding,
 - painting and coating,
 - gluing and sealing,
 - laser and water jet cutting,
 - milling and drilling,
 - casting and die spraying,
 - deburring and grinding,
 - screwing, wiring and fastening,
 - assembly of mechanical and electrical groups,
 - assembly of electronic boards.

- Besides material handling and manipulation, in a manufacturing process it is necessary to perform measurements to test product quality.
- The robot's capability to explore 3D space together with the availability of measurements on the manipulator's status allow a robot to be used as a measuring device.
- Typical applications include:
 - object inspection,
 - contour finding,
 - detection of manufacturing imperfections.

Video Video2

Applications

Advanced Robotics

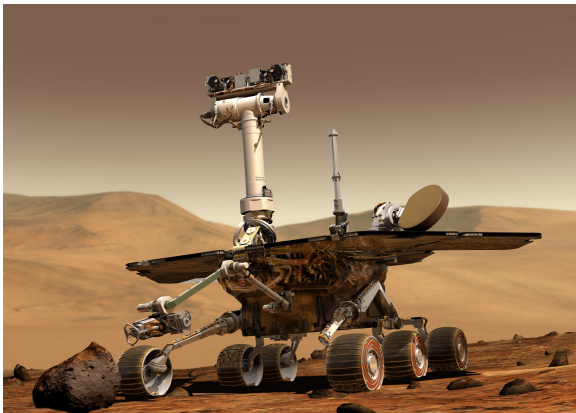
- The expression advanced robotics usually refers to the science studying robots with marked characteristics of autonomy, operating in scarcely structured or unstructured environments, whose geometrical or physical characteristics would not be known a priori.
- There are many motivations which strongly encourage advances in knowledge within this field.
- They range from the need for automata whenever human operators are not available or are not safe (**field robots**), to the opportunity of developing products for potentially wide markets which are aimed at improving quality of life (**service robots**).

- The context is that of deploying robots in areas where human beings could not survive or be exposed to unsustainable risks.
- Such robots should carry out exploration tasks and report useful data on the environment to a remote operator, using suitable onboard sensors.
- Typical scenarios are the exploration of a volcano, the intervention in areas contaminated by poisonous gas or radiation, or the exploration of the deep ocean or space.
- A similar scenario is that of disasters caused by fires in tunnels or earthquakes; in such occurrences, there is a danger of further explosions, escape of harmful gases or collapse, and thus human rescue teams may cooperate with robot rescue teams.

Video

Mars Rover

- NASA succeeded in delivering some mobile robots (rovers) to Mars which navigated on the Martian soil, across rocks, hills and crevasses. Such rovers were partially teleoperated from earth and have successfully explored the environment with sufficient autonomy.



Service Robots - 1

- Autonomous vehicles are also employed for civil applications, i.e., for mass transit systems, thus contributing to the reduction of pollution levels.
- Such vehicles are part of the so-called Intelligent Transportation Systems (ITS) devoted to traffic management in urban areas.



Video

Service Robots - 2

- Another feasible application where the adoption of mobile robots offers potential advantages is museum guided tours.



Service Robots - 3

- Technology is ready to transform into commercial products the prototypes of robotic aids to enhance elderly and impaired people's autonomy in everyday life; autonomous wheelchairs, mobility aid lifters, feeding aids and rehabilitation robots allowing tetraplegics to perform manual labor tasks are examples of such service devices.



Service Robots - 4

- Several robotic systems are employed for medical applications.
- Surgery assistance systems exploit a robot's high accuracy to position a tool, i.e., for hip prosthesis implant.
- Yet, in minimally-invasive surgery, i.e., cardiac surgery, the surgeon operates while seated comfortably at a console viewing a 3D image of the surgical field, and operating the surgical instruments remotely by means of a haptic interface.



Service Robots - 5

- Agricultural robotics are used to improve productivity, specialization and environmental sustainability.
- Labor shortages, increased consumer demand and high production costs are some of the factors that have accelerated automation in this sector, with the aim of reducing costs and optimizing harvests.



Video Video2

Service Robots - 6

- Another wide market segment comes from entertainment, where robots are used as toy companions for children, and life companions for the elderly, such as humanoid robots and the pet robots being developed in Japan.



Robot Modeling, Planning and Control

- In all robot applications, completion of a generic task requires the execution of a specific motion prescribed to the robot.
- The correct execution of such motion is entrusted to the control system which should provide the robot's actuators with the commands consistent with the desired motion.
- Motion control demands an accurate analysis of the characteristics of the mechanical structure, actuators, and sensors.
- The goal of such analysis is the derivation of the mathematical models describing the input/output relationship characterizing the robot components.
- Modelling a robot manipulator is therefore a necessary premise to finding motion control strategies.

Planning: robot manipulators

- With reference to the tasks assigned to a manipulator, the issue is whether to specify the motion at the joints or directly at the end-effector.
- In material handling tasks, it is sufficient to assign only the pick-up and release locations of an object (point-to-point motion)
- In machining tasks, the end-effector has to follow a desired trajectory (path motion).
- The goal of trajectory planning is to generate the timing laws for the relevant variables (joint or end-effector) starting from a concise description of the desired motion.

Video

Planning: mobile robots

- The motion planning problem for a mobile robot concerns the generation of trajectories to take the vehicle from a given initial configuration to a desired final configuration.
- Such a problem is more complex than that of robot manipulators, since trajectories have to be generated in respect of the kinematic constraints imposed by the wheels.
- Whenever obstacles are present in a mobile robot's workspace, the planned motions must be safe, so as to avoid collisions.
- Such a problem, known as motion planning, can be formulated in an effective fashion for both robot manipulators and mobile robots utilizing the configuration space concept.

Video

Control: robot manipulators

- Realization of the motion specified by the control law requires the employment of actuators and sensors.
- The trajectories generated constitute the reference inputs to the motion control system of the mechanical structure.
- The problem of robot manipulator control is to find the time behaviour of the forces and torques to be delivered by the joint actuators so as to ensure the execution of the reference trajectories.
- This problem is quite complex, since a manipulator is an articulated system and, as such, the motion of one link influences the motion of the others.

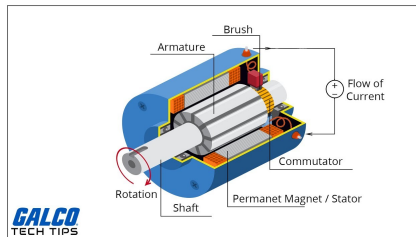
- Control of a mobile robot substantially differs from the analogous problem for robot manipulators.
- This is due, in turn, to the availability of fewer control inputs than the robot has configuration variables.
- An important consequence is that the structure of a controller allowing a robot to follow a trajectory (tracking problem) is unavoidably different from that of a controller aimed at taking the robot to a given configuration (regulation problem).
- Further, since a mobile robot's proprioceptive sensors do not yield any data on the vehicle's configuration, it is necessary to develop localization methods for the robot in the environment.

Actuators and Sensors

- Electric motors are used to “actuate” some parts of your robot: its wheels, legs, tracks, arms, fingers, or camera.
- There are dozens of types of electric motors but we focus on the most common types used in robotics:
 - Brushed DC motor
 - Brushless DC motor
 - Geared DC motor
 - Servo motor
 - Stepper motor
 - DC Linear Actuator
- AC (alternating current) motors are rarely used in mobile robots because most of the robots are powered with direct current (DC) coming from batteries.
- AC motors are mainly used in industrial environments where very high torque is required, or where the motors are connected to the mains / wall outlet

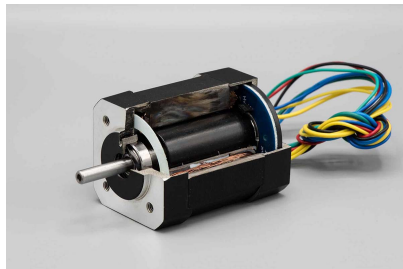
Brushed DC Motor

- A brushed DC electric motor is an internally commutated electric motor designed to be run from a direct current power source and utilizing an electric brush for contact.
- Advantages:
 - Inexpensive
 - Lightweight
 - Reasonably Efficient
 - Good low-speed torque
- Limitations:
 - In addition to the audible whine from the commutator brushes, these motors create a lot of electrical noise which can find its way back into other circuitry and cause problems.



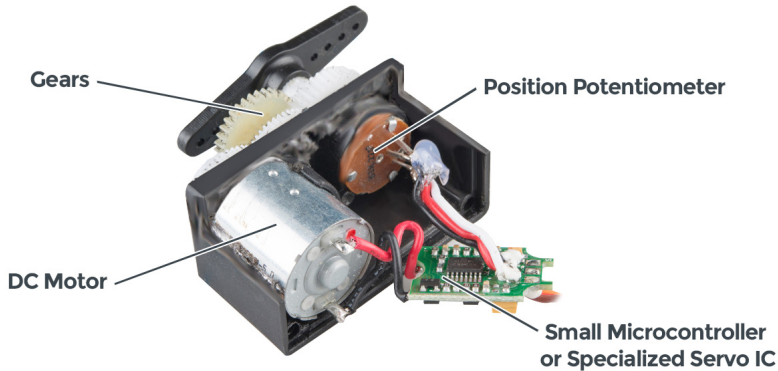
Brushless DC Motor

- A brushless DC electric motor is a synchronous motor using a direct current (DC) electric power supply.
- Advantages:
 - Higher torque to weight ratio
 - Reduced operational and mechanical noise
 - Longer lifespan (no brush and commutator erosion)
- Limitations:
 - Some types of brushless motors require a separate controller for operation.



Servo Motor

- A servomotor is an actuator that allows for precise control of position, velocity, and acceleration.
- It consists of a suitable motor coupled to a sensor for position feedback.
- It also requires a controller.



Stepper Motor

- A brushless DC electric motor is a synchronous motor using a direct current (DC) electric power supply.
- Advantages:
 - Precise repeatable positioning
 - Precise speed control
 - Excellent low-speed torque
 - Excellent 'holding torque' to maintain position
- Limitations:
 - Low efficiency
 - May need encoder or limit switch to establish a reference position
 - Subject to missed steps if overloaded

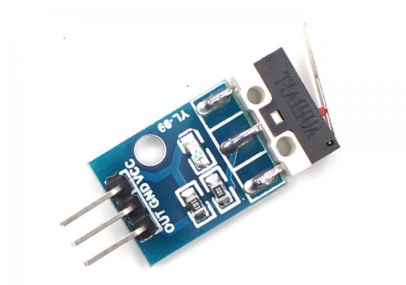


Collision sensors

- Switch sensor
 - Very simple
 - On-Off

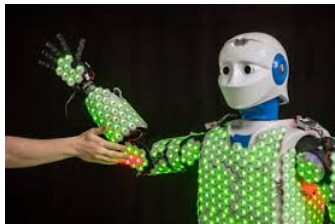
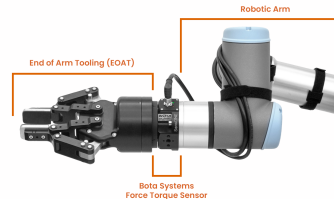


- 3D collision sensor



Interaction sensors

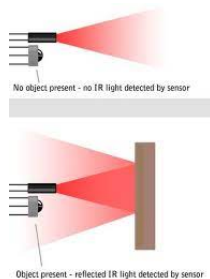
- Force/torque sensors
 - 3-axis force sensor
 - 3-axis torque sensor



- Artificial skin
 - Temperature
 - Pressure
 - Proximity

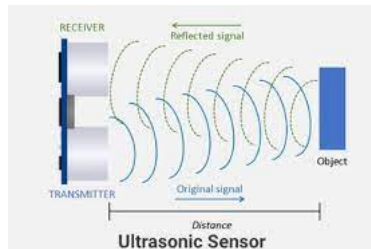
Proximity/distance sensors - 1

- Infrared (IR) sensor
 - Measures reflected light intensity which corresponds to distance
 - Range: 10 cm – 1 m
 - Wide detection cone
 - Can be also used to detect black vs white color (e.g., for a line detection)
 - Unaffected by material softness
 - Sensitive to object color and transparency
 - Relatively short distance



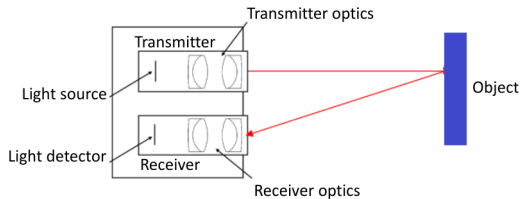
Proximity/distance sensors - 2

- Ultrasonic sensor
 - Measures time of reflected signal to return which corresponds to distance
 - Range: 2 cm – 3 m
 - Wide detection cone
 - Unaffected by object color and transparency
 - Sensitive to material softness
 - Relatively slow



Proximity/distance sensors - 3

- LIDAR sensor (laser imaging, detection, and ranging)
 - Measures time of reflected signal to return which corresponds to distance
 - Range: 5 cm – 40 m
 - Fast and long range sensing
 - One single point (no cone)
 - Relatively expensive (1D)



- LIDAR 2D

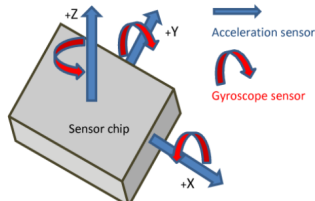
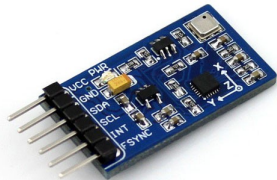


- LIDAR 3D



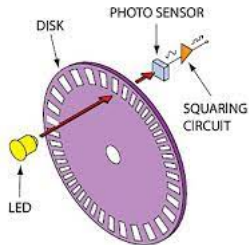
Inertial motion sensors

- Inertial motion unit (IMU) sensor
 - 3-axis gyroscope
 - 3-axis accelerometer
 - 3-axis magnetometer (absolute orientation)
 - Can be used for estimation of position of a mobile robot using path integration
Path reconstruction is inaccurate and gets worse with time



Position sensors

- Angular position sensor (optical encoder)
 - Measures rotary motion
 - Can be used for estimation of position of a mobile robot using path integration
 - Path reconstruction is inaccurate and gets worse with time



Localization sensors

- Global positioning system (GPS) sensor
 - Global position
 - Requires satellite signal
 - Works only outdoors



- Indoor positioning system (IPS)
 - Radio frequency (RF) signals from radio sources to determine robots location

2D cameras

- 2D RGB cameras
 - Direct mapping of environment
 - 2D RGB image
 - Produces a lot of data: $3 \times W \times H$

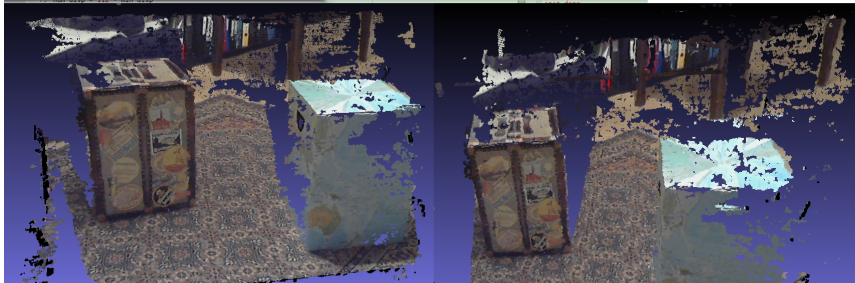
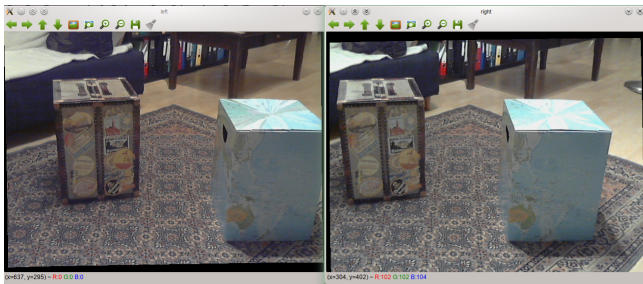


- 3D RGB-D sensors
 - RGB camera
 - IR camera (depth sensor)
 - Allows 3D reconstruction
 - Sensitive to object transparency
 - Produces a lot of data: $3 \times W \times H + 1 \times W \times H$



- 3D stereo cameras
 - Two RGB cameras
 - Allows 3D reconstruction
 - Not affected by object transparency
 - Produces a lot of data: $2 \times 3 \times W \times H$

3D cameras - 2



- Video 3D Lidar
- Video RGBD
- Video RGBD + SLAM